

Intro to Open LLMs for the Behavioral Sciences

Dirk Wulff & Zak Hussain

@ SPUDM 2025



MAX PLANCK INSTITUTE
FOR HUMAN DEVELOPMENT



Goals

Familiarize you with the workings and applications of open-source LLMs and how to implement them using the Hugging Face ecosystem



Software stack



+



+



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<> Code

About

dwulff

Update README.md

8858b35 · 3 days ago

🕒 30 Commits

📁 day_1	add day_1.ipynb	2 weeks ago
📁 day_2	add data	last week
📁 day_3	typo	last week
📄 .gitignore	init	4 months ago
📄 LICENSE.txt	init	4 months ago
📄 README.md	Update README.md	3 days ago
📄 cover.png	init	4 months ago

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👁 1 watching

🔗 0 forks

Releases

No releases published

Create a new release

Packages

No packages published

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✎

☰

Schedule

Day 1

09:30 AM - 10:15 AM: Welcome & Intro

10:15 AM - 11:15 AM: [Talk: Intro to LLMs](#)

11:15 AM - 11:30 AM: Break

11:30 AM - 12:30 PM: Discussion: Find applications in small groups

12:30 PM - 02:00 PM: Lunch

02:00 PM - 02:30 PM: [Talk: A gentle intro to Hugging Face and Python](#)

02:30 PM - 02:45 PM: Setup Colab

02:45 PM - 03:15 PM: [Exercise: Running pipelines](#)

03:15 PM - 03:30 PM: Break

03:30 PM - 03:45 PM: Walkthrough

03:45 AM - 04:30 AM: [Talk: Intro to transformers & embeddings](#)

04:30 PM - 05:00 PM: Open questions

Day 2

09:30 AM - 10:00 AM: Recap quiz

10:00 AM - 11:00 AM: [Talk: Intro to transformers & embeddings \(continued\)](#)

11:00 AM - 11:15 AM: Break

11:15 AM - 12:00 AM: [Exercise: Embeddings for clustering](#)

12:00 PM - 12:30 PM: Walkthrough

12:30 PM - 02:00 PM: Lunch

02:00 PM - 03:00 PM: Discussion: Find applications in small groups

03:00 PM - 03:30 PM: Break

03:30 PM - 04:30 PM: [Intro to textgen](#)

04:30 PM - 05:00 PM: Open questions

Day 3

09:30 AM - 10:00 AM: Recap quiz

10:00 AM - 11:00 AM: Exercise: Text gen for [classification](#)

11:00 AM - 11:30 AM: Walkthrough

11:30 PM - 11:45 PM: Break

11:45 PM - 12:30 PM: Discussion: Find applications in small groups



30s each

1. Where are you from, what is your background, and what do you do?
2. How much experiences do you have with machine learning?
3. What motivates you to learn more about LLMs?

Intro LLMs

Dirk Wulff & Zak Hussain

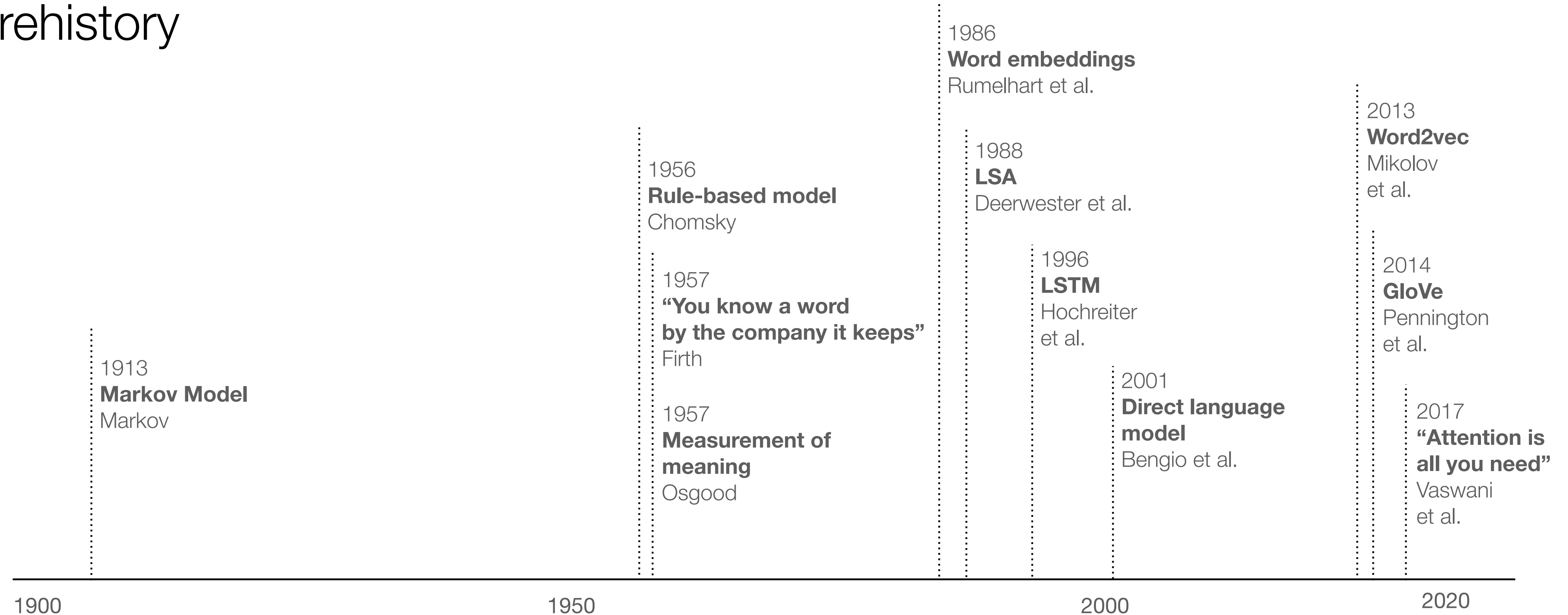


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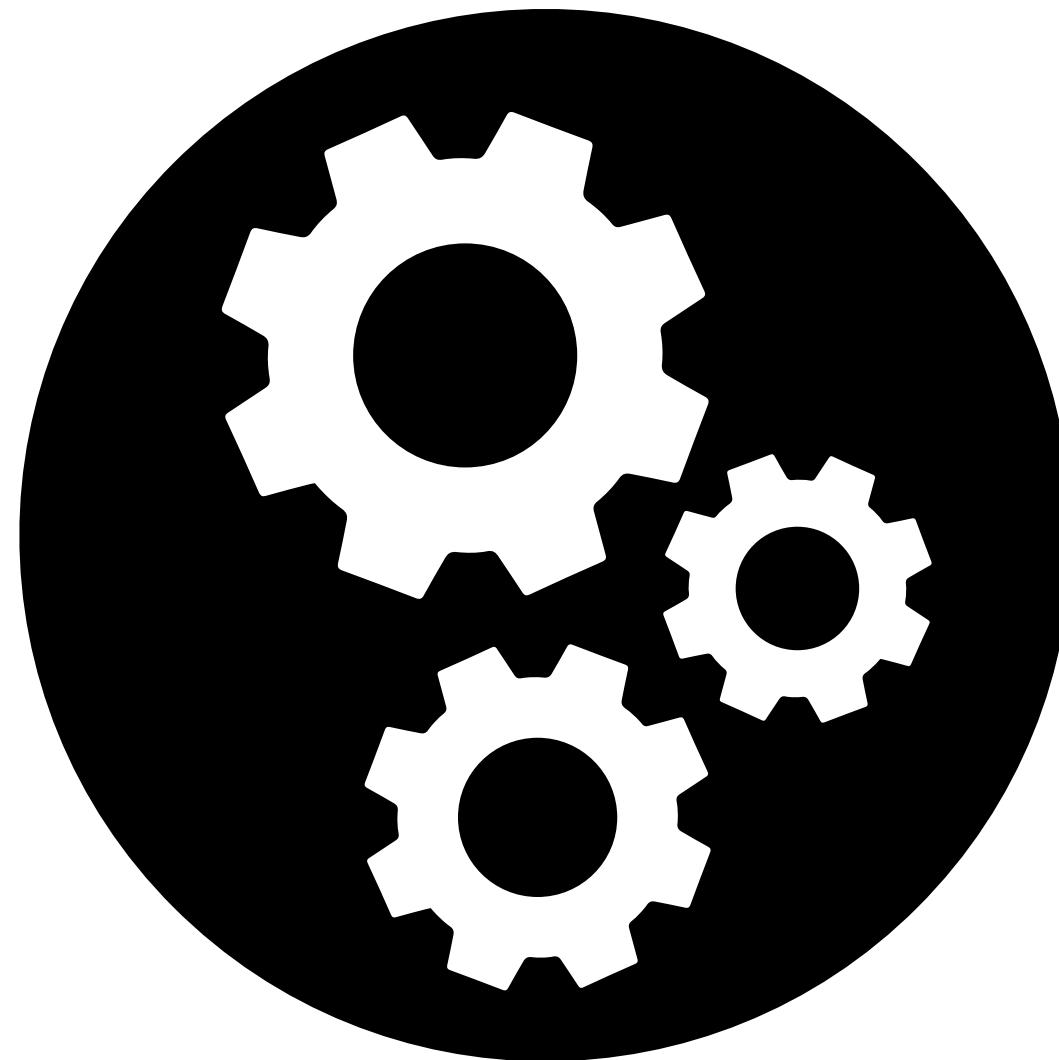
Language models

Prehistory



LLMs as mechanisms

LLM



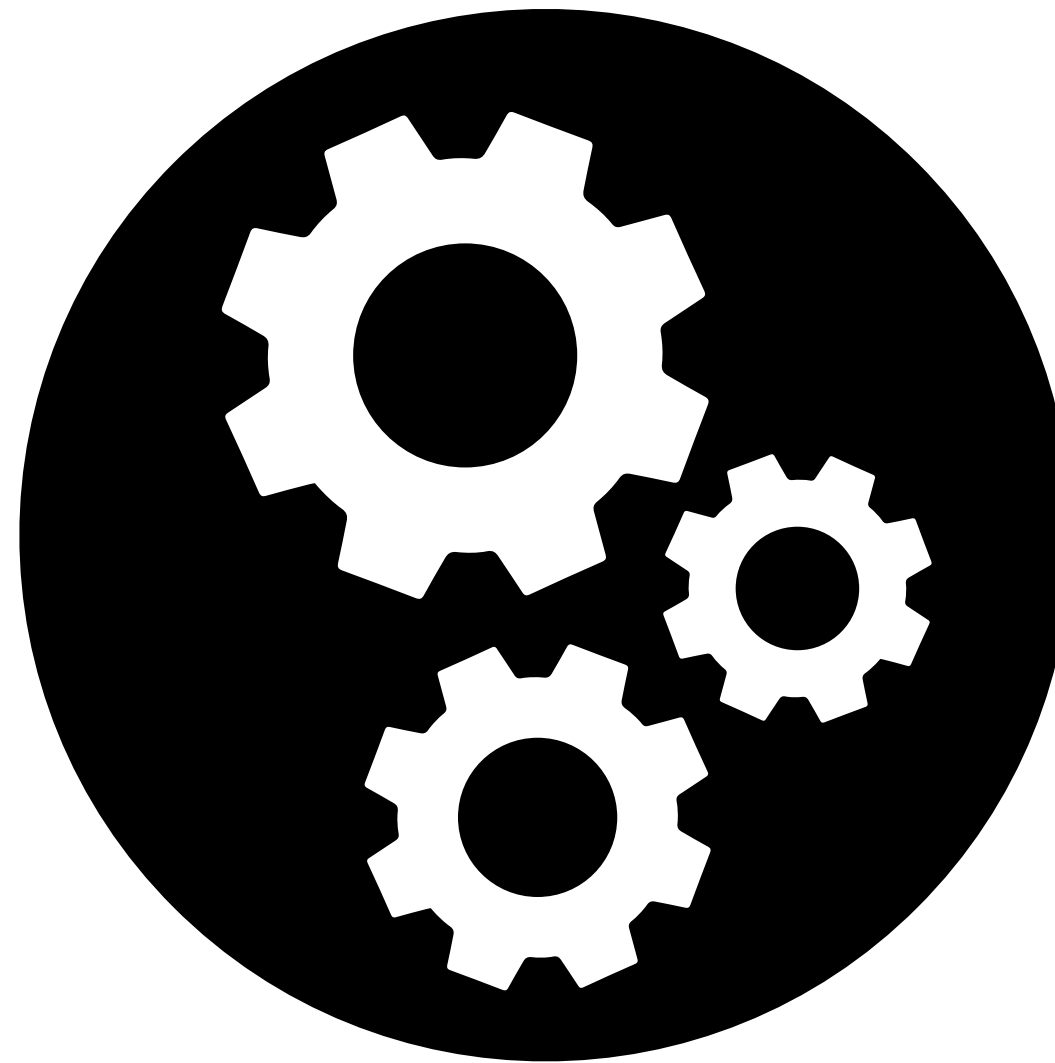
LLMs as mechanisms

Input

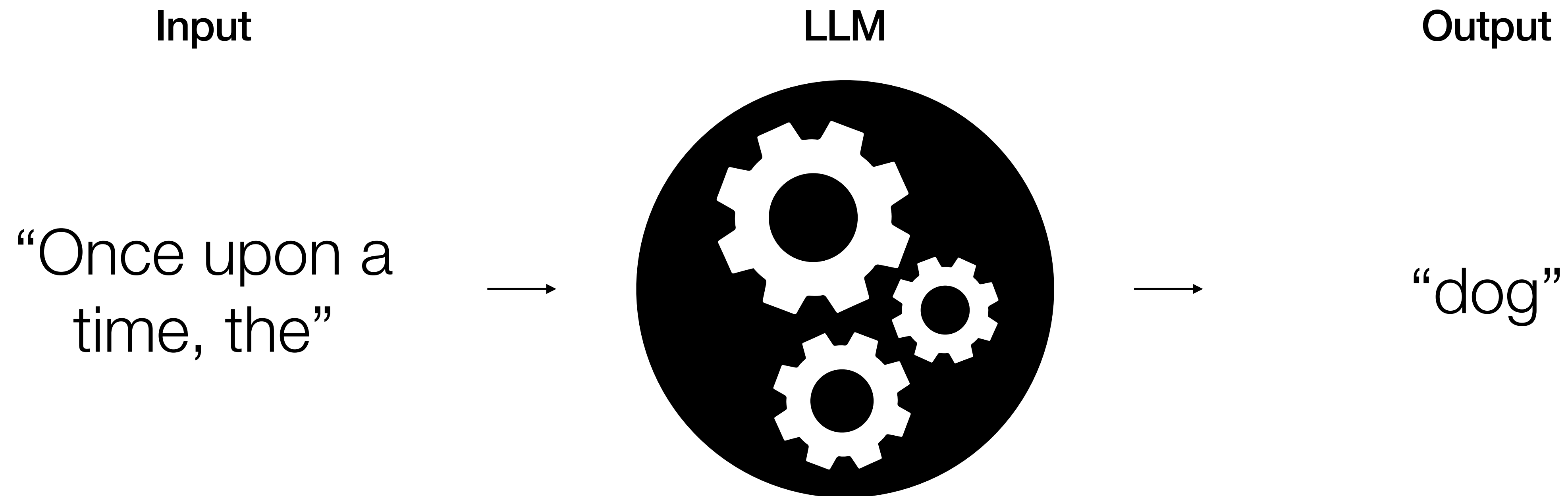
“Once upon a
time, the”



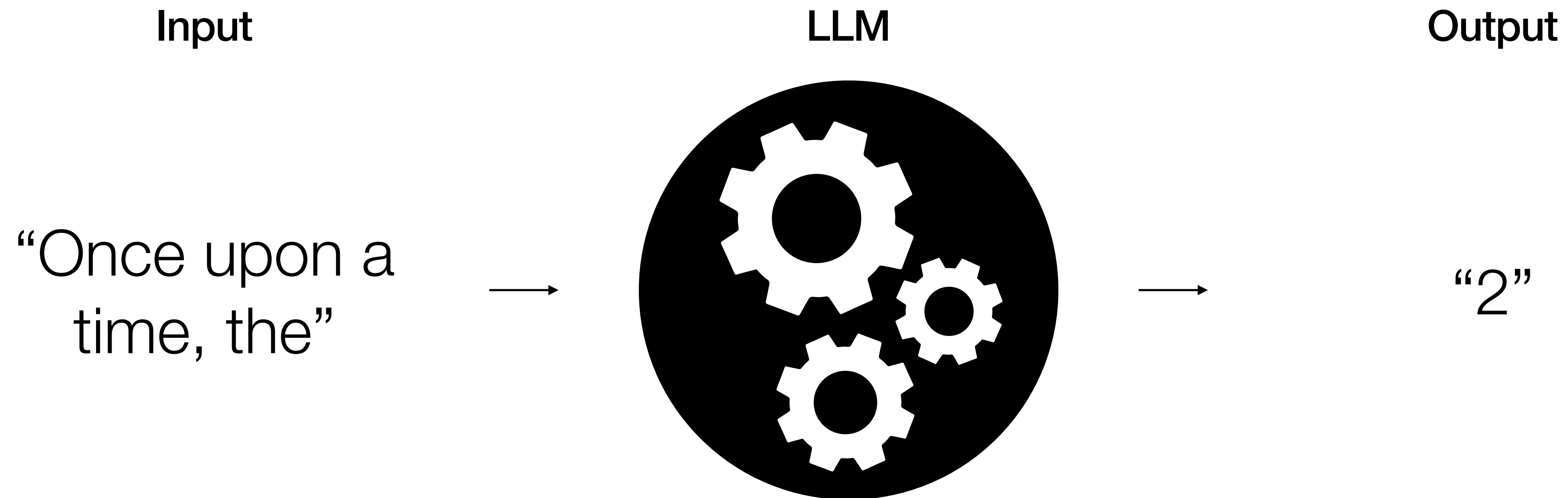
LLM



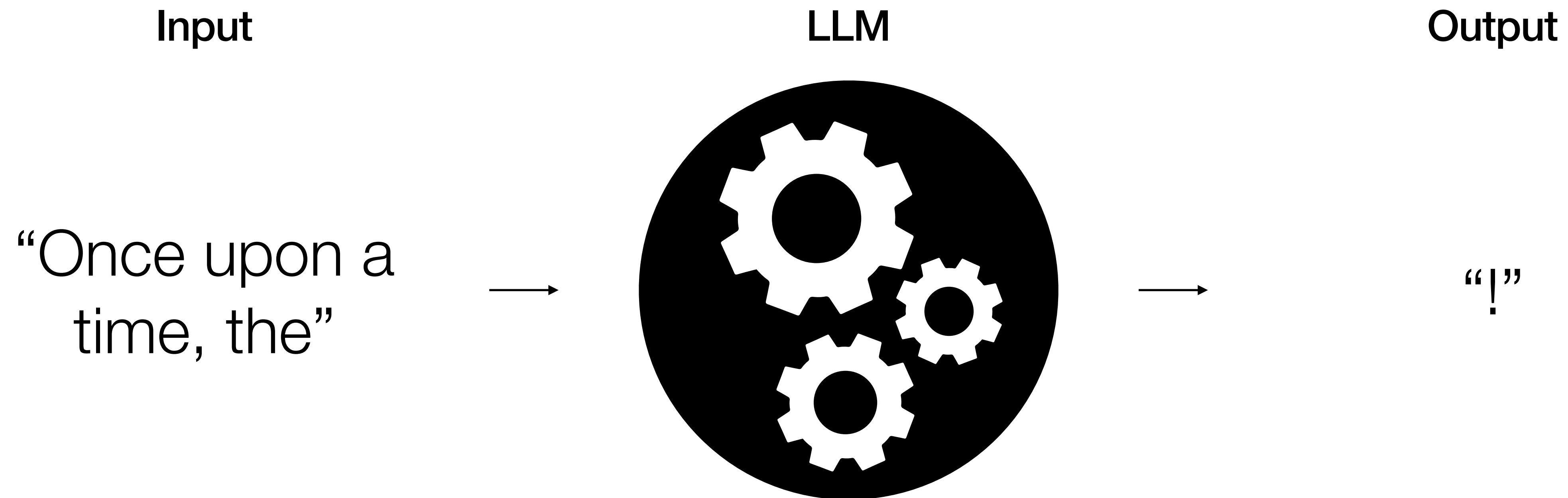
LLMs as mechanisms



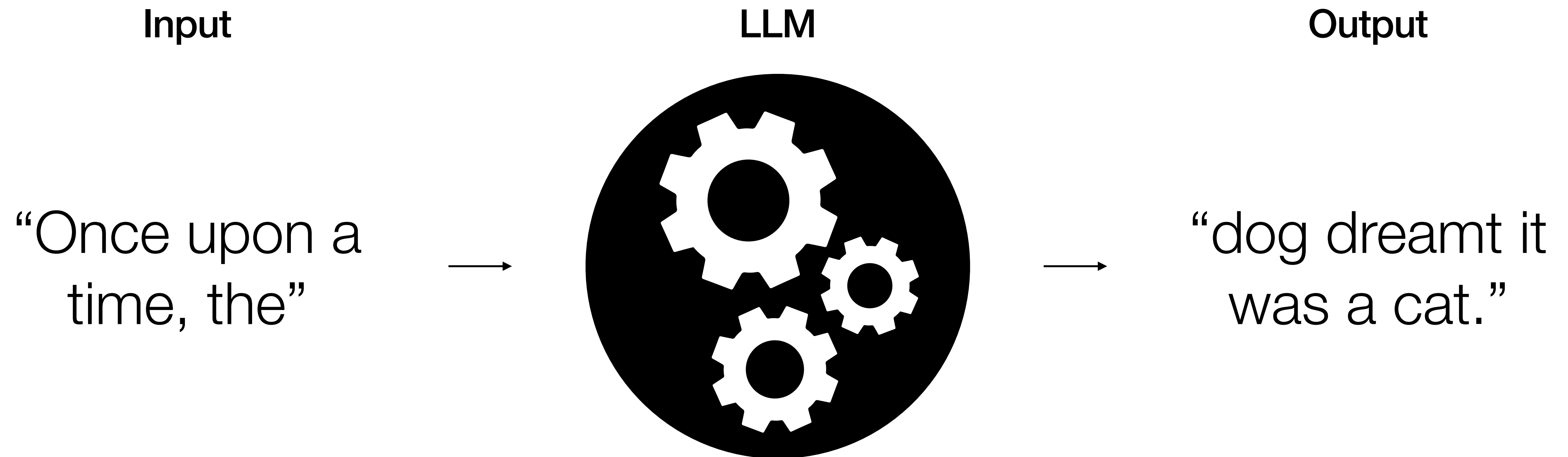
LLMs as mechanisms



LLMs as mechanisms



LLMs as mechanisms



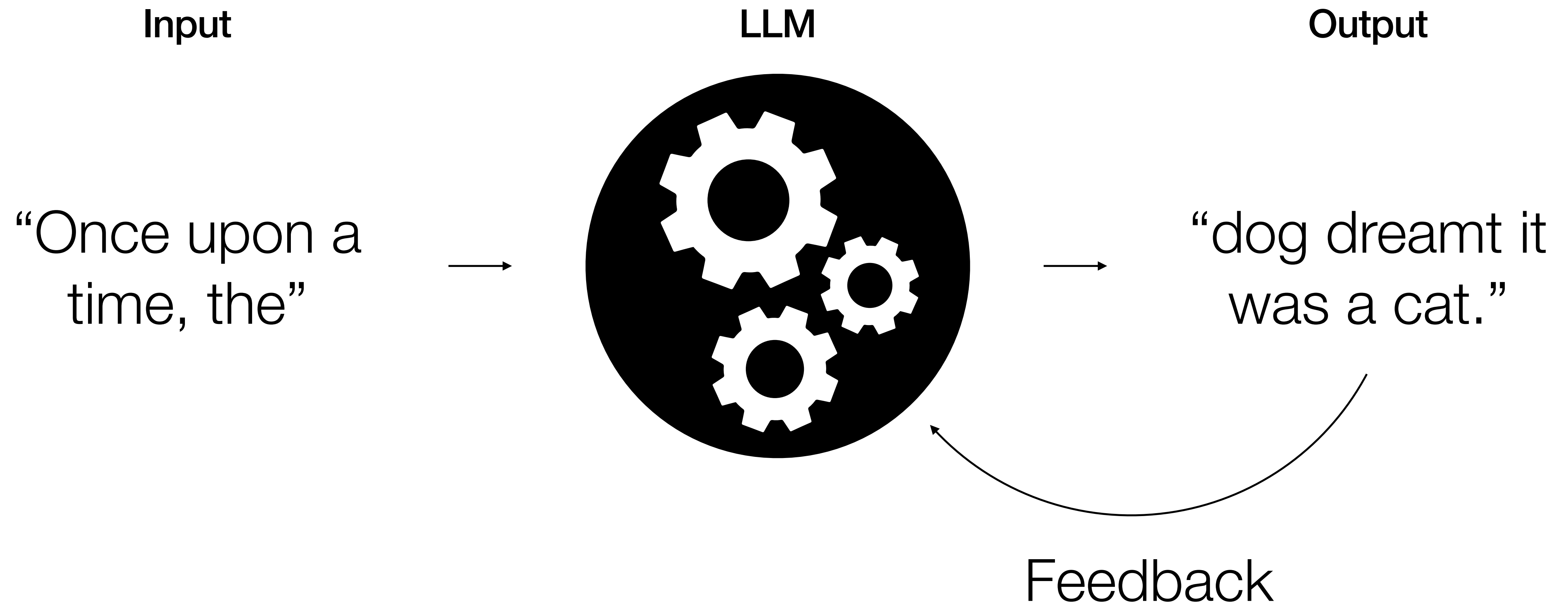
Once upon a time, the



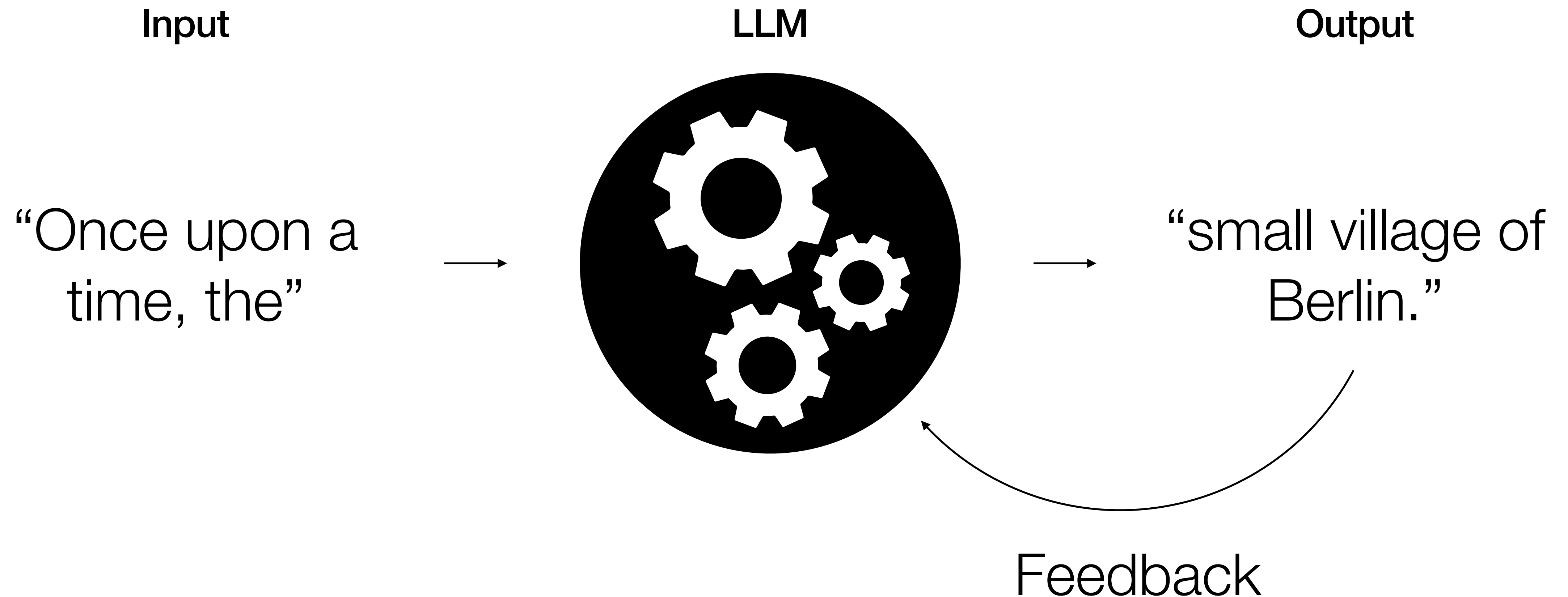
Once upon a time, the small village of Elmswood was nestled in a lush valley surrounded by towering mountains. The villagers lived peacefully, their days marked by the rhythms of nature and the changing seasons. However, everything changed when a mysterious old man arrived, carrying with him a locked chest that was said to contain a secret capable of altering the course of history. Intrigued by the stranger and his enigmatic treasure, the people of Elmswood soon found themselves on the brink of an adventure that would bind them together in ways they could never have imagined.



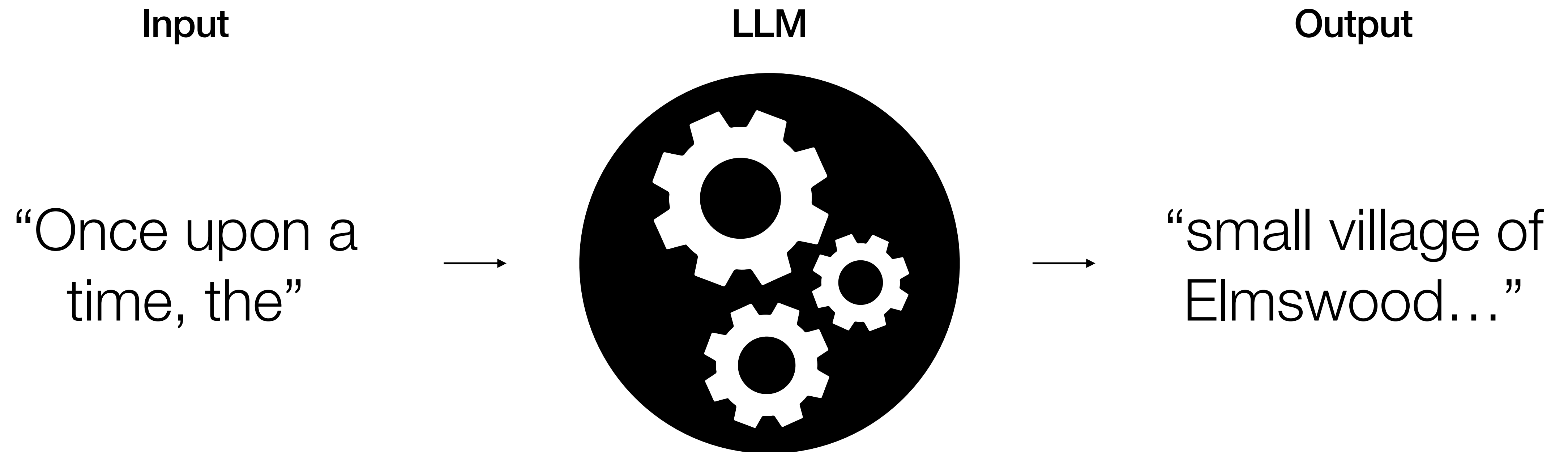
LLMs as **trained** mechanisms



LLMs as **trained** mechanisms



LLMs as **trained** mechanisms



Phi-3-mini-4K-Instruct and Phi-3-mini-128K-Instruct were trained over 7 days on 3.3T tokens using 512 H100-80G GPUs for each model. They followed advanced fine-tuning techniques to align with human preferences and safety standards.

The pre-training process followed two distinct and consecutive stages:

- *In the first stage*, the models were primarily exposed to a vast collection of web sources. This data helped the models develop general knowledge and language comprehension.
- *In the second stage*, the models were fine-tuned with a more rigorously selected subset of web data from the first phase, combined with additional synthetic data, to improve their logical reasoning and specialized abilities.

After these 2 stages, the models underwent *additional training*, which included supervised instruction fine-tuning and preference tuning, to enhance their stability and security.

The training dataset, made of 3.3 trillion tokens, is a meticulously curated mix of quality-filtered public documents, select educational materials, code, and newly generated synthetic data generated by LLMs. Specifically, the team filtered the web data to encompass the appropriate degree of *knowledge* and retained a greater number of web pages that may enhance the models' reasoning abilities. Instead of indiscriminately feeding vast amounts of data into the training model, the emphasis was placed on enhancing its reasoning capabilities, rather than one that merely has a vast repository of information.



Phi-3

LLM training = Pretraining + Fine-tuning

Trillions of tokens
Millions of power
consumption
Uses masked/next token
prediction

Hand-selected/crafted texts
Quality input-output pairs
Human feedback

Masked/next token prediction

"Once upon a time" is a [stock phrase](#) used to introduce a narrative of past events, typically in [fairy tales](#) and folk tales. It has been used in some form since at least 1380 (according to the [Oxford English Dictionary](#)) in [storytelling](#) in the [English language](#) and has started many narratives since 1600. These stories sometimes end with "and they all lived [happily ever after](#)", or, originally, "happily until their deaths".

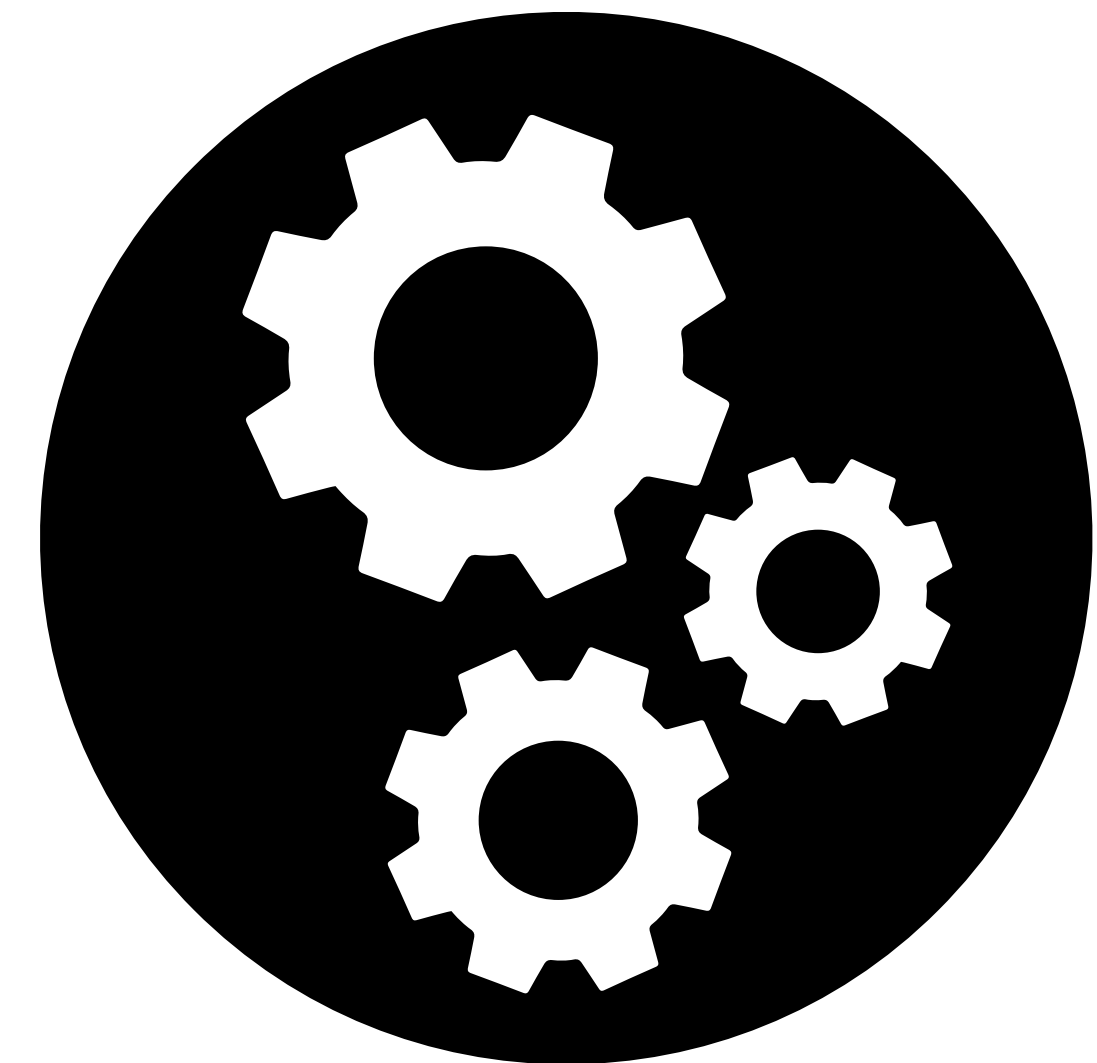
The phrase is common in [fairy tales](#) for younger children. It was used in the original translations of the stories of [Charles Perrault](#) as a translation for the [French](#) "*il était une fois*", of [Hans Christian Andersen](#) as a translation for the [Danish](#) "*der var engang*" (literally "there was once"), the [Brothers Grimm](#) as a translation for the [German](#) "*es war einmal*" (literally "it was once") and [Joseph Jacobs](#) in [English](#) translations and fairy tales.

In *More English Fairy Tales*, Joseph Jacobs notes that:

"The opening formula are varied enough, but none of them has much play of fancy. 'Once upon a time and a very good time it was, though it wasn't in my time nor in your time nor in any one else's time.' is effective enough for a fairy epoch, and is common, according to Mayhew (London Labour, III), among tramps."^[1]

https://en.wikipedia.org/wiki/Once_upon_a_time

LLM



Masked/next token prediction

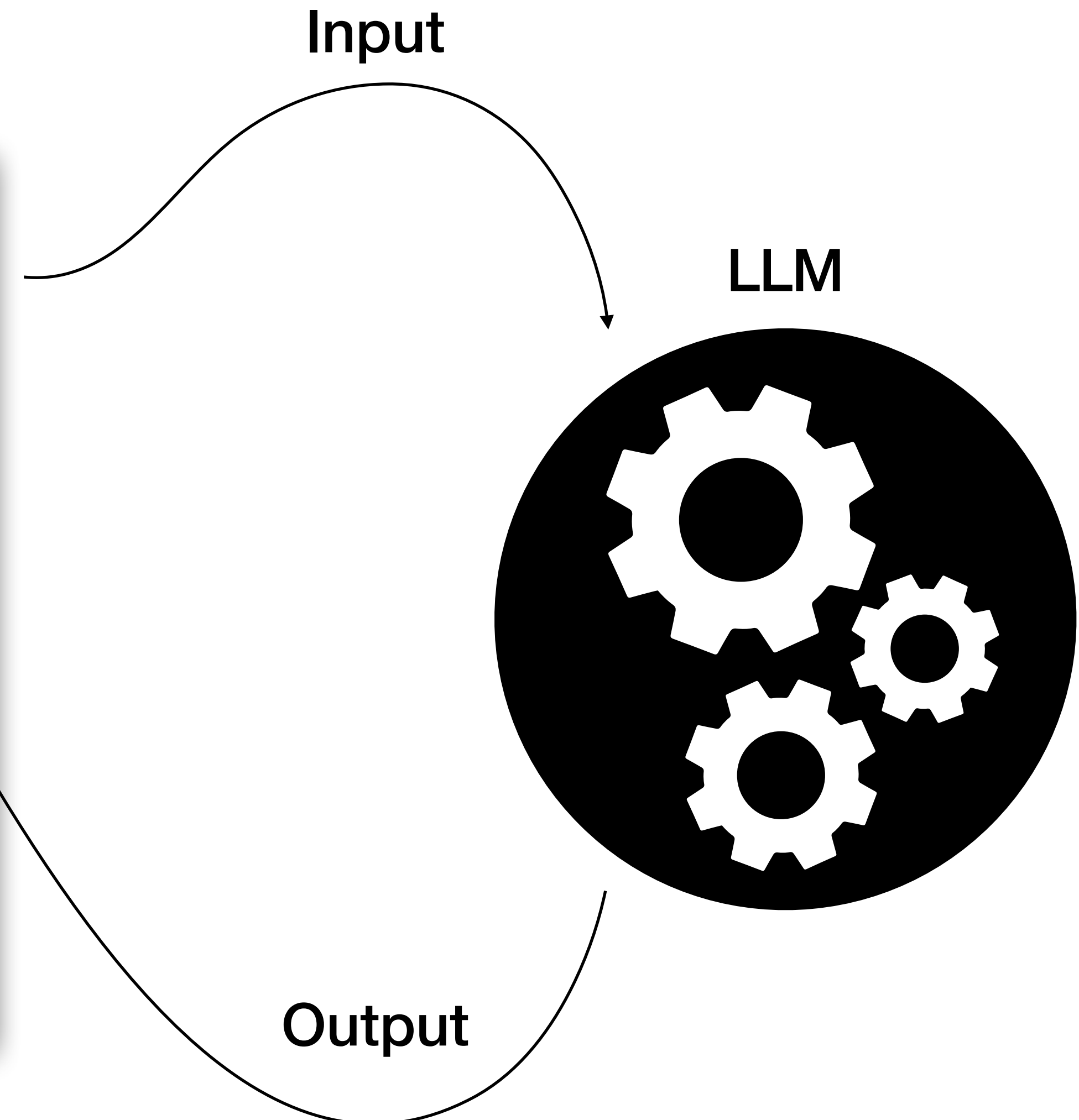
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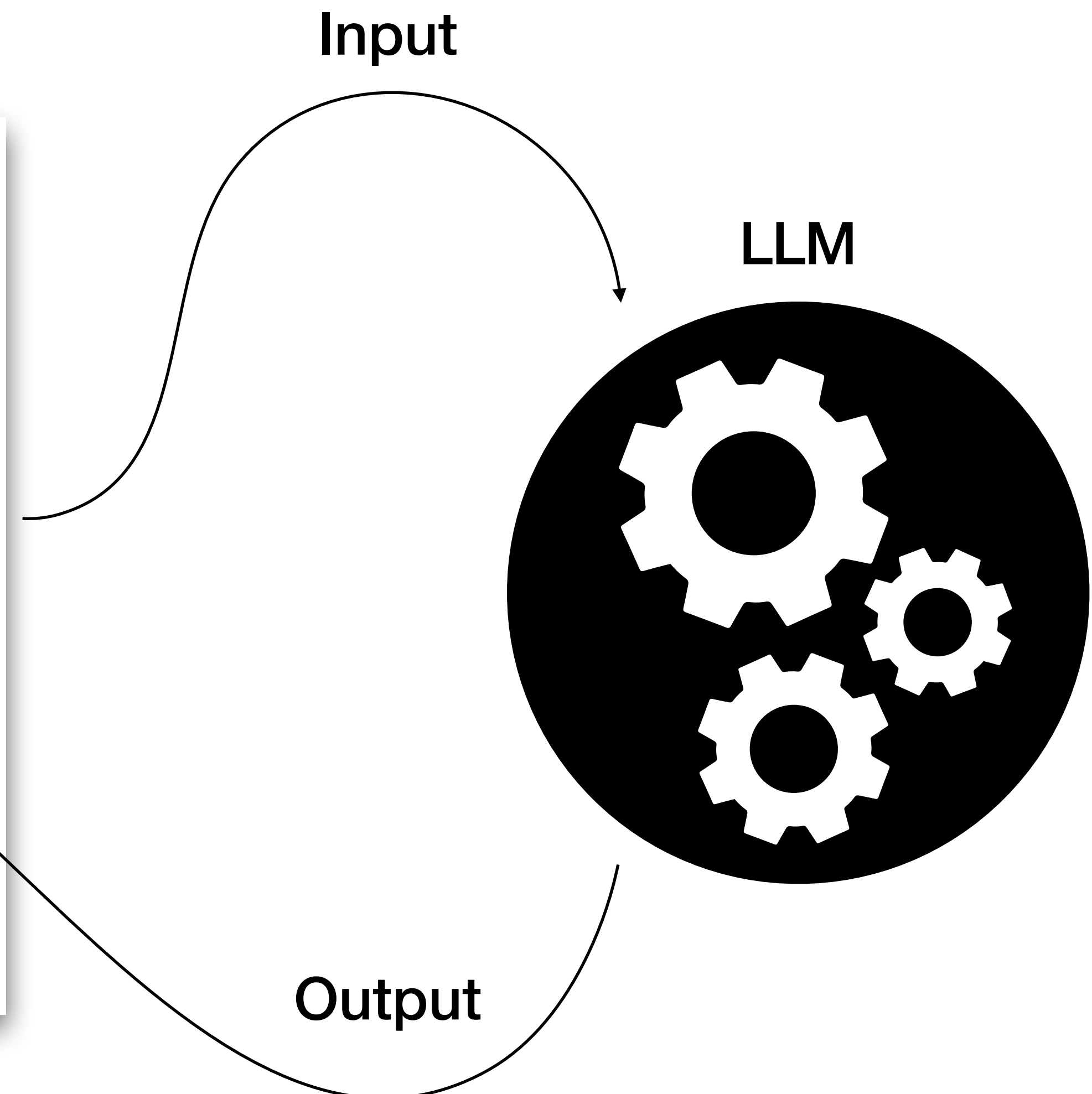
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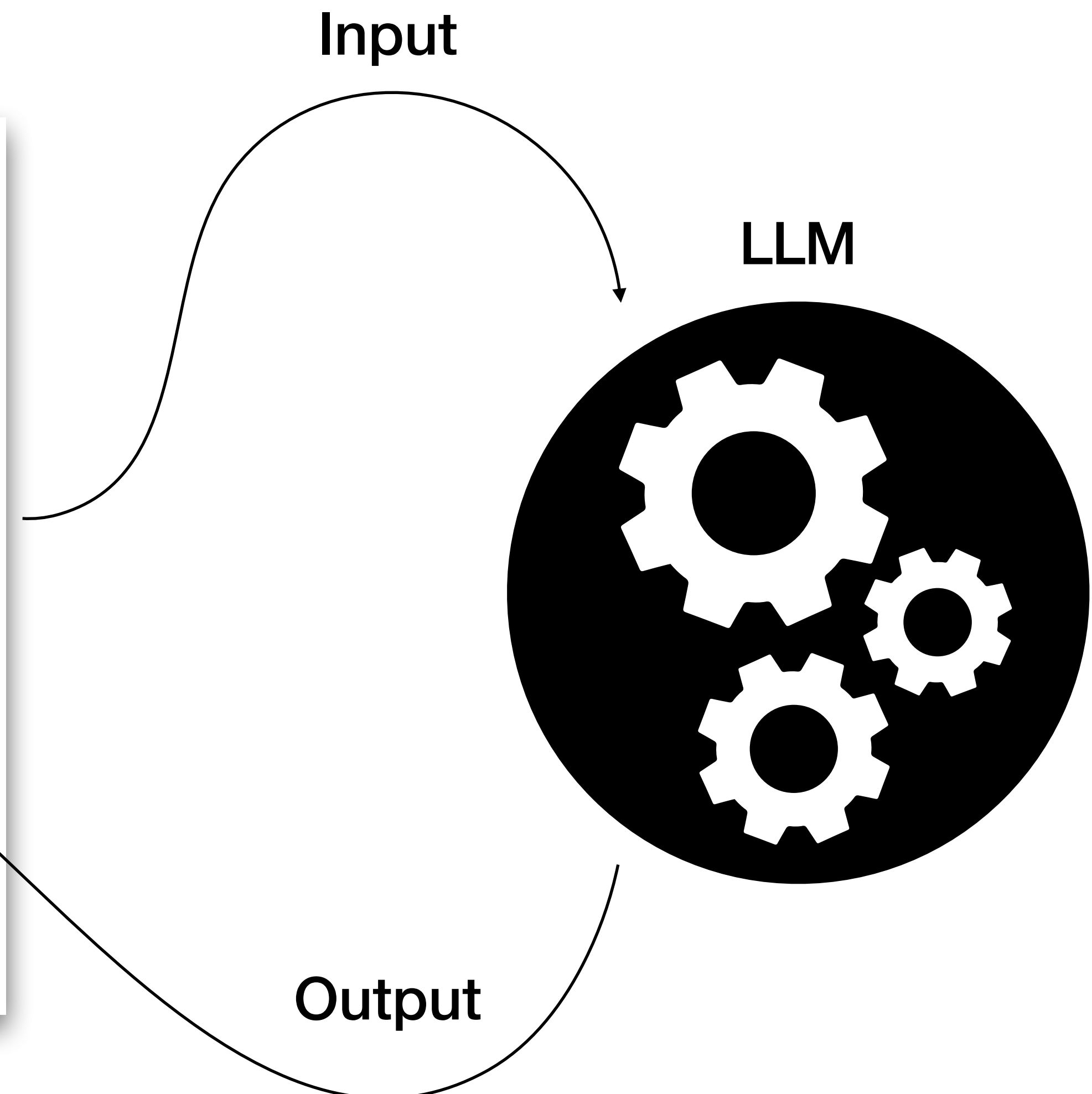
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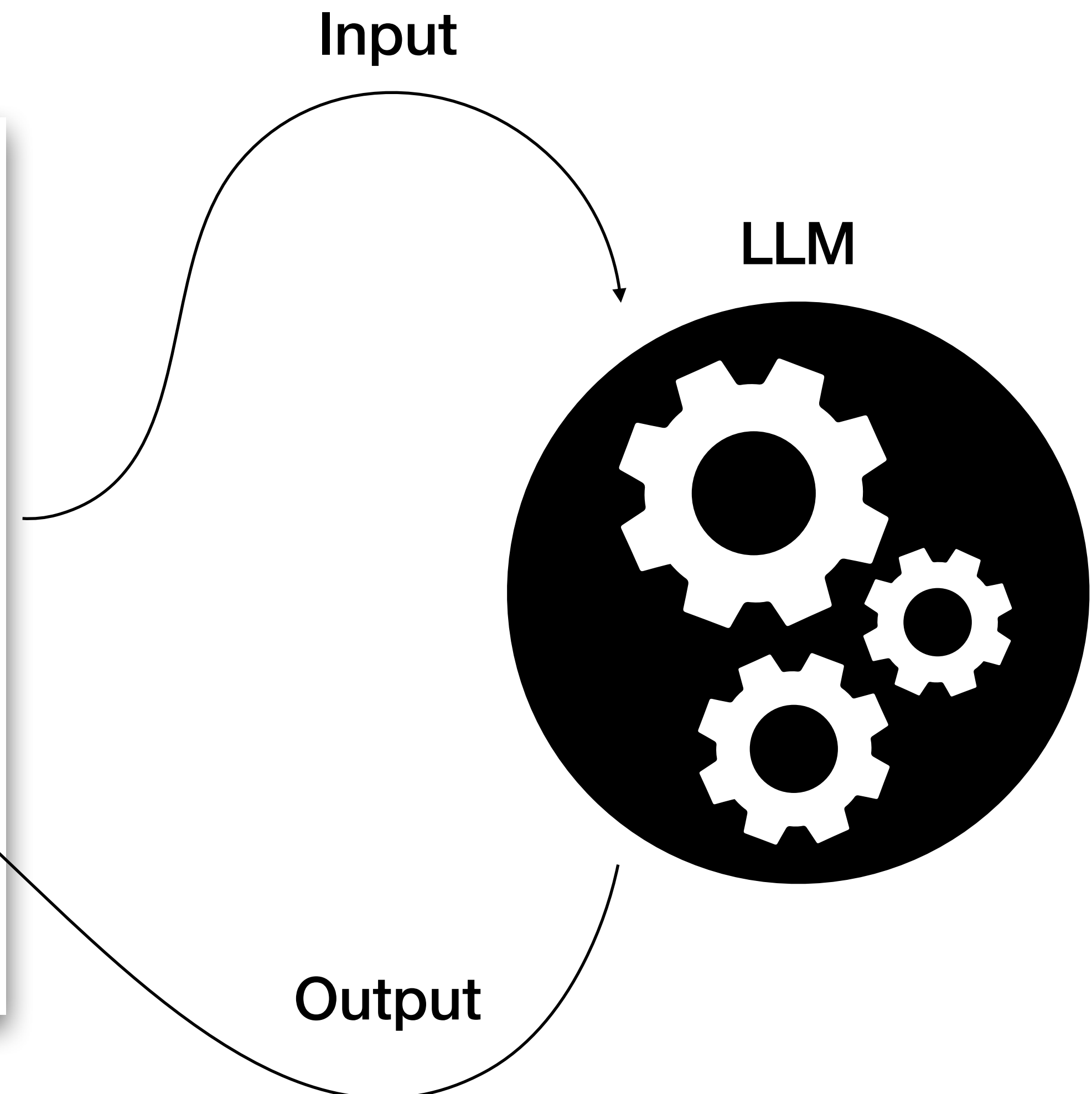
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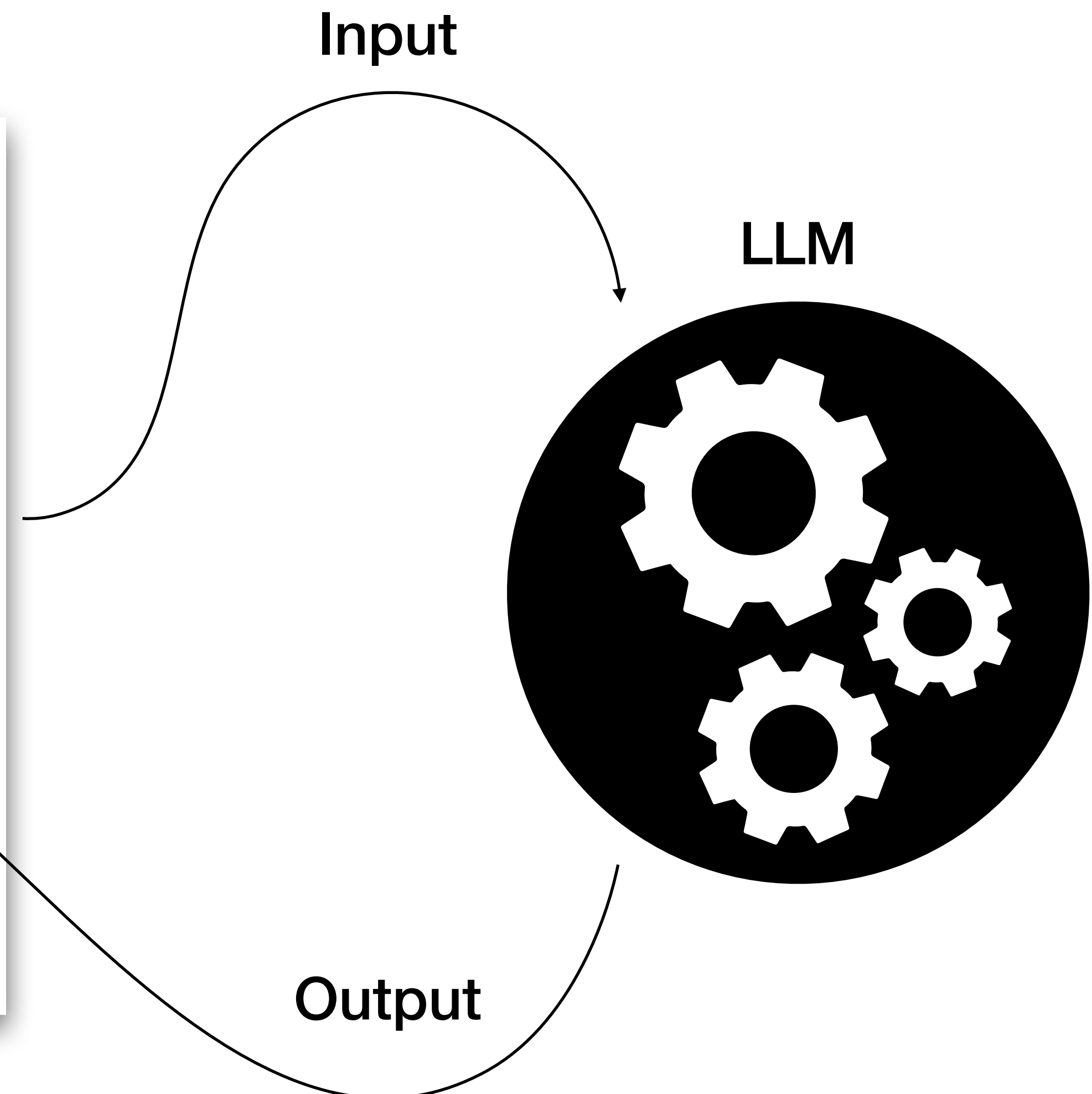
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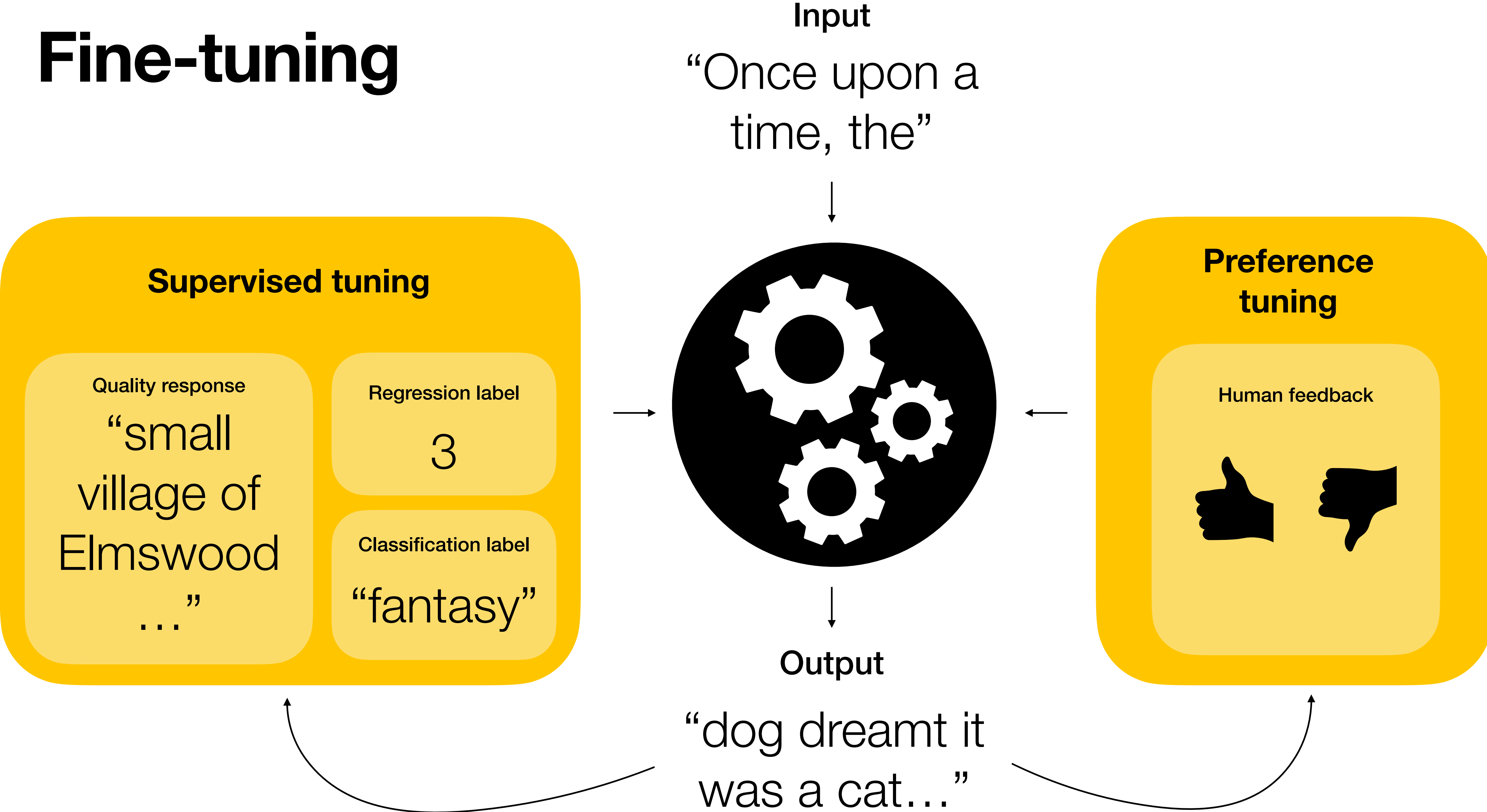
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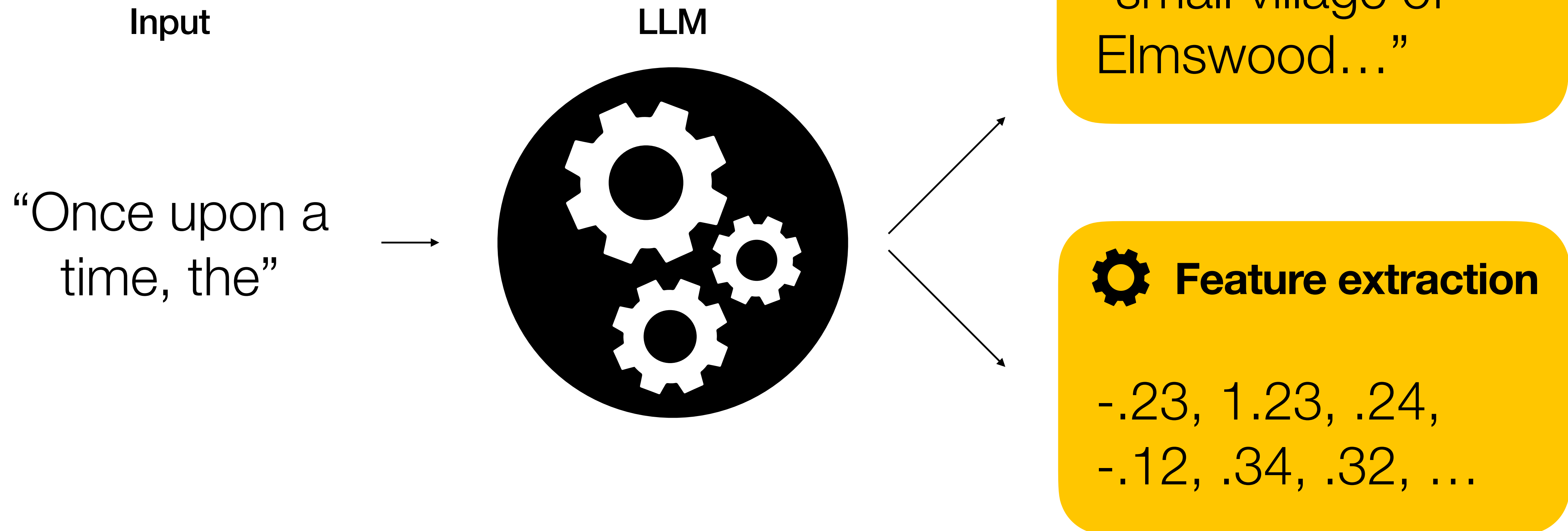
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Fine-tuning



Two major applications



Closed vs. Open

Public use

Performance

Safety

Reproducible

Data secure

Interpretable



Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

Audio-Text-to-Text Image-Text-to-Text

Visual Question Answering

Document Question Answering Video-Text-to-Text

Visual Document Retrieval Any-to-Any

Computer Vision

Depth Estimation Image Classification

Object Detection Image Segmentation

Text-to-Image Image-to-Text Image-to-Image

Image-to-Video Unconditional Image Generation

Video Classification Text-to-Video

Zero-Shot Image Classification Mask Generation

Zero-Shot Object Detection Text-to-3D

Image-to-3D Image Feature Extraction

Keypoint Detection

Natural Language Processing

Text Classification Token Classification

Table Question Answering Question Answering

Zero-Shot Classification Translation

Summarization Feature Extraction

Models 1,680,396

Filter by name

Full-text search

Sort: Trending

nvidia/parakeet-tdt-0.6b-v2

Automatic Speech Recognition • Updated 10 days ago • 72.7k • 691

nari-labs/Dia-1.6B

Text-to-Speech • Updated 5 days ago • 148k • 2.04k

JetBrains/Mellum-4b-base

Text Generation • Updated 4 days ago • 2.81k • 311

deepseek-ai/DeepSeek-Prover-V2-671B

Text Generation • Updated 11 days ago • 7.38k • 750

black-forest-labs/FLUX.1-dev

Text-to-Image • Updated Aug 16, 2024 • 2.66M • 10.1k

Qwen/Qwen3-30B-A3B

Text Generation • Updated 11 days ago • 150k • 514

tencent/HunyuanCustom

Image-to-Video • Updated 2 days ago • 77

hexgrad/Kokoro-82M

Text-to-Speech • Updated about 1 month ago • 1.88M • 4.27k

ServiceNow-AI/Apriel-Nemotron-15b-Thinker

Updated 4 days ago • 784 • 66

ACE-Step/ACE-Step-v1-3.5B

Text-to-Audio • Updated 1 day ago • 310

Lightricks/LTX-Video

Text-to-Video • Updated 5 days ago • 214k • 1.38k

Qwen/Qwen3-235B-A22B

Text Generation • Updated 10 days ago • 82.4k • 764

lodestones/Chroma

Text-to-Image • Updated 1 day ago • 401

cognition-ai/Kevin-32B

Updated 5 days ago • 342 • 93

microsoft/Phi-4-reasoning-plus

Text Generation • Updated 3 days ago • 10.8k • 240

Tesslate/UIGEN-T2-7B-Q8_0-GGUF

Text Generation • Updated 5 days ago • 2.59k • 115

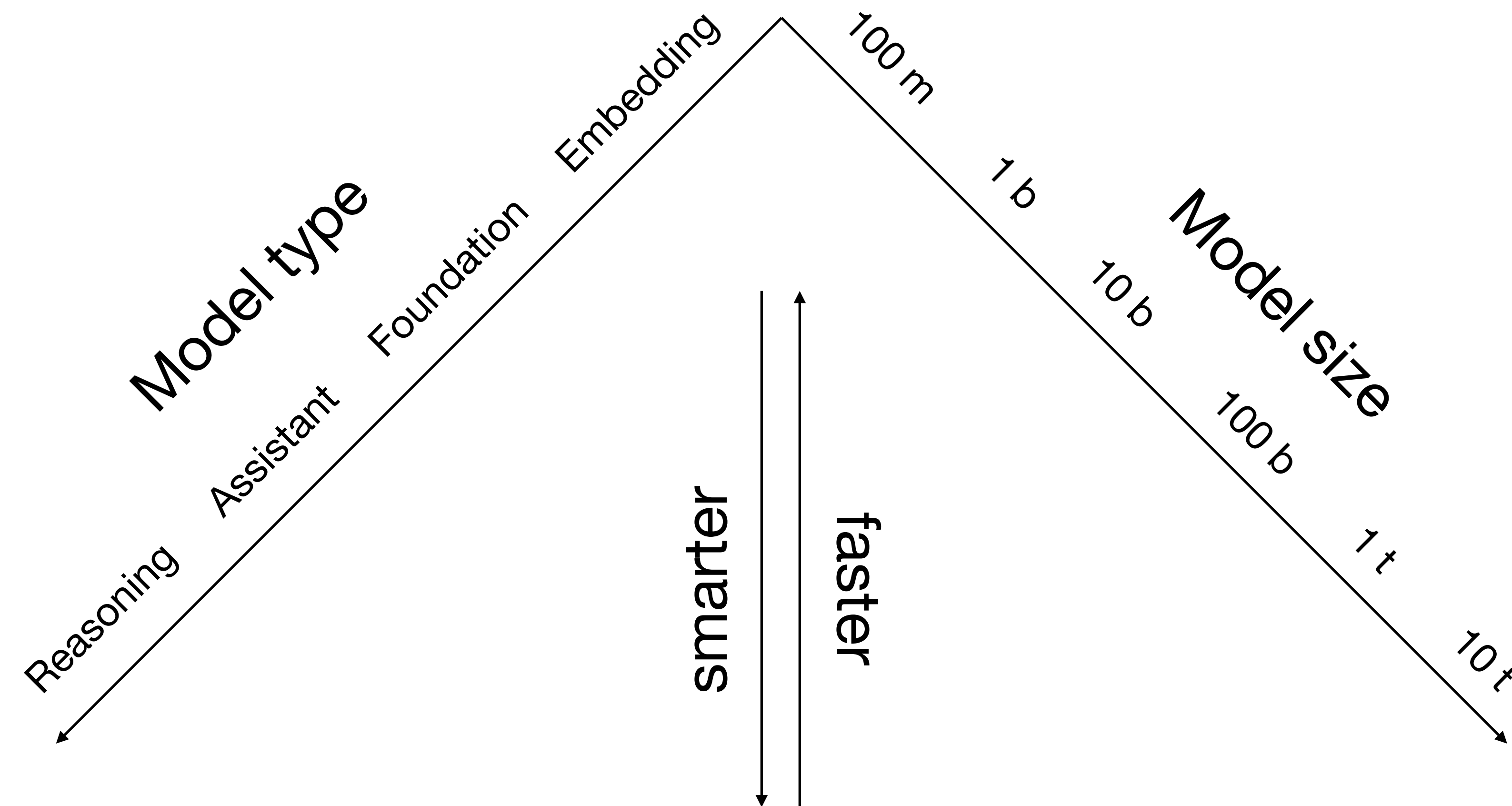
fdtn-ai/Foundation-Sec-8B

Text Generation • Updated 10 days ago • 21.5k • 145

Qwen/Qwen3-8B

Text Generation • Updated 12 days ago • 394k • 262

Distinguishing LLMs



Project	Availability						Documentation						Access		
(maker, bases, URL)	Open code	LLM data	LLM weights	RL data	RL weights	License	Code	Architecture	Preprint	Paper	Modelcard	Datasheet	Package	API	
OLMo 7B Instruct	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	~	
Ai2	LLM base: OLMo 7B			RL base: OpenInstruct											12.5
BLOOMZ	✓	✓	✓	✓	~	~	✓	✓	✓	✓	✓	✓	✗	✓	
bigscience-workshop	LLM base: BLOOMZ, mT0			RL base: xP3											12.0
AmberChat	✓	✓	✓	✓	✓	✓	~	~	✓	✗	~	~	✗	✓	
LLM360	LLM base: Amber			RL base: ShareGPT + Evol-Instruct (sy...											10.0
Open Assistant	✓	✓	✓	✓	✗	✓	✓	✓	~	✗	✗	✗	✓	✓	
LAION-AI	LLM base: Pythia 12B			RL base: OpenAssistant Conversations											9.5
...															
Command R+	✗	✗	✗	✓	✓	~	✗	✗	✗	✗	~	✗	✗	✗	
Cohere AI	LLM base:			RL base: Aya Collection											3.0
LLaMA2 Chat	✗	✗	~	✗	~	✗	✗	~	~	✗	~	✗	✗	~	
Facebook Research	LLM base: LLaMA2			RL base: Meta, StackExchange, Anthropic											3.0
Nanbeige2-Chat	✓	✗	✗	✗	✓	~	✗	✗	✗	✗	✗	✗	✗	~	
Nanbeige LLM lab	LLM base: Unknown			RL base: Unknown											3.0
Llama 3 Instruct	✗	✗	~	✗	~	✗	✗	~	✗	✗	~	✗	✗	~	
Facebook Research	LLM base: Meta Llama 3			RL base: Meta, undocumented											2.5
Solar 70B	✗	✗	~	✗	~	✗	✗	✗	✗	✗	~	✗	✗	~	
Upstage AI	LLM base: LLaMA2			RL base: Orca-style, Alpaca-style											2.0
Xwin-LM	✗	✗	~	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	~	
Xwin-LM	LLM base: LLaMA2			RL base: unknown											1.0
ChatGPT	✗	✗	✗	✗	✗	✗	✗	✗	~	✗	✗	✗	✗	✗	
OpenAI	LLM base: GPT 3.5			RL base: Instruct-GPT											0.5

see also <https://osai-index.eu/the-index>

LMSYS Chatbot Arena Leaderboard

[Vote](#) | [Blog](#) | [GitHub](#) | [Paper](#) | [Dataset](#) | [Twitter](#) | [Discord](#) |

LMSYS [Chatbot Arena](#) is a crowdsourced open platform for LLM evals. We've collected over **1,000,000** human pairwise comparisons to rank LLMs with the [Bradley-Terry model](#) and display the model ratings in Elo-scale. You can find more details in our [paper](#).

Arena

Full Leaderboard

Total #models: **99**. Total #votes: **1,170,955**. Last updated: 2024-05-20.

 **NEW!** View leaderboard for different categories (e.g., coding, long user query)! This is still in preview and subject to change.

Rank* (UB) ▲	Rank (StyleCtrl) ▲	Model ▲	Arena Score ▲	95% CI ▲	Votes ▲	Organization	License ▲
1	1	Gemini-Exp-1206	1374	+4/-5	20227	Google	Proprietary
1	1	ChatGPT-4o-latest (2024-11-20)	1365	+4/-3	33383	OpenAI	Proprietary
1	4	Gemini-2.0-Flash-Thinking-Exp-1219	1364	+5/-6	15728	Google	Proprietary
2	4	Gemini-2.0-Flash-Exp	1357	+6/-4	19030	Google	Proprietary
3	1	o1-2024-12-17	1351	+7/-7	7289	OpenAI	Proprietary
6	4	o1-preview	1335	+4/-4	33194	OpenAI	Proprietary
7	7	DeepSeek-V3	1319	+6/-6	10510	DeepSeek	DeepSeek
7	10	Step-2-16K-Exp	1305	+8/-9	3374	StepFun	Proprietary

Overall

Bitext Mining

Classification

Clustering

Pair Classification

Reranking

Retrieval

STS

Summarization

Retrieval w/Instructions

English

Chinese

French

Polish

Overall MTEB English leaderboard 🏆

○ Metric: Various, refer to task tabs

○ Languages: English

Rank ▲	Model ▲	Model Size (Million Parameters) ▲	Memory Usage (GB, fp32) ▲	Embedding Dimensions ▲	Max Tokens ▲	Average (56 datasets) ▲	Classification Average (12 datasets) ▲	Clustering Average (11 datasets) ▲
1	NV-Embed-v1					69.32	87.35	52.8
2	voyage-large-2-instruct			1024	16000	68.28	81.49	53.35
3	SFR-Embedding-Mistral	7111	26.49	4096	32768	67.56	78.33	51.67
4	gte-Qwen1.5-7B-instruct	7099	26.45	4096	32768	67.34	79.6	55.83
5	voyage-lite-02-instruct	1220	4.54	1024	4000	67.13	79.25	52.42
6	GritLM-7B	7242	26.98	4096	32768	66.76	79.46	50.61
7	e5-mistral-7b-instruct	7111	26.49	4096	32768	66.63	78.47	50.26
8	google-gecko.text-embedding-p	1200	4.47	768	2048	66.31	81.17	47.48
9	SE_v1					65.66	76.8	47.38
10	GritLM-8x7B	46703	173.98	4096	32768	65.66	78.53	50.14

Models

BERTish

all-MiniLM-L6-v2
MPNet
MPNet-personality



Feature extraction
(fine-tuning)

Open GPTs

Llama 3.2 3B



Text-generation
(in-context learning)

Quiz



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Quiz

How are LLMs trained?

What are the main ways to use LLMs?

What aspects distinguish different LLMs?

How to choose an LLM?